

Matrix inequalities and norms

Constant one in the logarithmic commutator inequality

Difficulty: challenging **Importance:** broadly interesting

Status: Open

Rating rationale: Improving the known universal coefficient to its proposed sharp value is challenging; connections between matrix logarithms, entropy and quantum dynamics make the question broadly interesting.

Projection equivalence and coefficient sharpness — 12 September 2026

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[Theorems 1 and 2](#) prove equivalence with the arbitrary-rank projection case and give a strictly positive order-two family showing that any universal coefficient is at least one. The proof allows component traces to vary and uses entropy concavity to recombine them. The universal coefficient-one upper bound remains Open.

A separate [independent Codex AI-agent audit](#) passed this limited scope. This is informal automated review; no complete resolution, historical novelty, external human peer review or formal verification is claimed. No Lean verification was performed.

Problem statement

For every integer $n \geq 1$ and positive definite $A, B \in \mathbb{C}^{n \times n}$ with $\text{tr}(A + B) = 1$, set $a = \text{tr } A$, $b = \text{tr } B$. Is

$$\|B \log(A + B) - \log(A + B)B\|_1 \leq -a \log a - b \log b$$

always true? The logarithm is the natural logarithm applied by spectral functional calculus; $\|X\|_1 = \text{tr}(X^*X)^{1/2}$. Thus $a, b > 0$ and $a + b = 1$, and every expression is well-defined.

The target is specifically the coefficient 1 on the right. Existence of some dimension-independent coefficient has been proved and is not the open question. The coefficient-one question is explicitly suggested immediately after the original displayed conjecture.

This is a sharp matrix-function estimate for the rate at which noncommuting positive matrices mix. It connects perturbation estimates for the matrix logarithm with entropy production and quantum information.

References

1. K. M. R. Audenaert and F. Kittaneh, *Problems and Conjectures in Matrix and Operator Inequalities*, arXiv:1201.5232v3 (2012), §6, Conjecture 4, equation (26), and the sentence proposing $c = 1$. [Full text](#).
2. K. M. R. Audenaert, *Quantum Skew Divergence*, J. Math. Phys. 55 (2014), 112202, §8, proof of small incremental mixing with coefficient 2. [Full text](#).
3. A. Vershynina, *Entanglement rates for Renyi, Tsallis and other entropies*, arXiv:1803.07117v3 (2021), §8, conclusion, pp. 15–16, explicitly distinguishing the proved coefficient 2 and proposed coefficient 1. [Full preprint](#).
4. Q. Ning, F.-Z. Guo, J. Zhang and Q.-Y. Wen, *On Bounding Entangling Rates and Mixing Rates in Some Special Cases*, Int. J. Theor. Phys. 55 (2016), 1686–1694. [Published abstract](#), which reports coefficient one under additional restrictions; the subscription body was not independently inspected in this audit.

Status check (2026-09-10): Audenaert's theorem settles the existence claim with 2. Vershynina's 2021 revision distinguishes the proved coefficient from the proposed coefficient 1. Searches for "small incremental mixing sharp constant", "logarithmic commutator Audenaert constant one", and 2025/2026 found later special-case estimates but no announced universal coefficient-one proof or counterexample. This limited search does not certify that the conjecture remains open.